

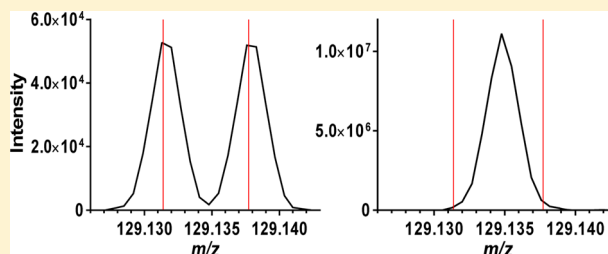
Ion Coalescence of Neutron Encoded TMT 10-Plex Reporter Ions

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S Supporting Information

ABSTRACT: Isobaric mass tag-based quantitative proteomics strategies such as iTRAQ and TMT utilize reporter ions in the low mass range of tandem MS spectra for relative quantification. The recent extension of TMT multiplexing to 10 conditions has been enabled by utilizing neutron encoded tags with reporter ion m/z differences of 6 mDa. The baseline resolution of these closely spaced tags is possible due to the high resolving power of current day mass spectrometers. In this work we evaluated the performance of the TMT10 isobaric mass tags on the Q Exactive Orbitrap mass spectrometers for the first time and demonstrated comparable quantification accuracy and precision to what can be achieved on the Orbitrap Elite mass spectrometers. However, we discovered, upon analysis of complex proteomics samples on the Q Exactive Orbitrap mass spectrometers, that the proximate TMT10 reporter ion pairs become prone to coalescence. The fusion of the different reporter ion signals into a single measurable entity has a detrimental effect on peptide and protein quantification. We established that the main reason for coalescence is the commonly accepted maximum ion target for MS2 spectra of 1e6 on the Q Exactive instruments. The coalescence artifact was completely removed by lowering the maximum ion target for MS2 spectra from 1e6 to 2e5 without any losses in identification depth or quantification quality of proteins.



In recent years mass spectrometry (MS)-based proteomics has progressed from a solely qualitative analysis to a robust quantitative description of the protein complement of cells and organisms.¹ The quantification methodologies are based on either label-free or stable isotope labeling (SIL) approaches.² This development has enabled the differential and dynamic analysis of proteins and their post-translational modifications in a wide range of applications including protein expression profiling,^{3–5} protein–protein interaction studies,^{6,7} cell signaling analysis,^{8–10} chemoproteomic target profiling,^{11–15} and biomarker studies.^{16–20} Isobaric (or tandem) mass tags²¹ such as iTRAQ²² and TMT²³ are popular peptide and protein labeling reagents in quantitative proteomics. The different isotope coded variants of each reagent produce isobarically labeled peptides that cannot be distinguished in MS1 scans. When the labeled peptides are fragmented in the mass spectrometer, a low molecular weight reporter ion is generated with a mass specific to the different isotope coded variants of the tag, and relative quantification can be achieved by comparing the reporter ion signals which correspond to the respective samples under investigation.

We and others have recently reported on a strategy to increase the multiplexing rate of TMT beyond 6-plex that utilizes the high resolving power of current mass spectrometry instrumentation to distinguish between reporter ions containing one extra neutron incorporated into either carbon or nitrogen.^{24,25} At a resolution of approximately 50,000, a similar precision and accuracy of quantification of these 6 mDa spaced reporter ions was achieved as with 1 Da spacing. 10-Plex TMT

reagents using this strategy have now become commercially available. Subsequently, others have extended the neutron encoding concept for increasing the multiplexing rates of metabolic and chemical label based MS1 quantification approaches.^{26,27}

Here we evaluate the performance of the TMT10 isobaric mass tags on the Q Exactive Orbitrap mass spectrometers for the first time and show that baseline resolution of 6 mDa spaced reporter ions and comparable quantification accuracy and precision can be achieved using this instrument family as when using the Orbitrap Elite.²⁸ However, we have found that abundant 6 mDa spaced reporter ions of the TMT10 tags are prone to coalesce^{29–31} when analyzing complex samples using the generally accepted instrument settings on the Q Exactive mass spectrometers. The coalescence phenomenon is well-known on FTICR instruments³¹ and has recently been reported for the first time on the Orbitrap Elite.³⁰ Phase locking of the ion clouds with close frequencies occurs when the ion populations exceed a certain threshold and makes it impossible to distinguish the two ion populations. Instead of two distinct closely spaced peaks, only a single peak will appear in the mass spectrum with an m/z value located between the true m/z values of the two ions. The coalescence of the two proximate reporter ion signals into a single entity had a detrimental effect on peptide quantification as well as on

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protein quantification in our experiments. Below we outline the extent of this effect on two different Q Exactive instruments and the Orbitrap Elite. We also evaluate and suggest instrument settings which completely remedy the coalescence problem without affecting either protein coverage or quantification precision.

MATERIALS AND METHODS

Sample Preparation for Mass Spectrometry. A 100 μ g portion of a tryptic digest from K562 cells lyzed in 8 M urea digested at 2 M were labeled using the following TMT10plex mass tags TMT126 (126.12772591 Th, $^{12}\text{C}_8\text{H}_{16}^{14}\text{N}_1^+$), TMT127L (127.12476080 Th, $^{12}\text{C}_8\text{H}_{16}^{15}\text{N}_1^+$), TMT127H (127.13108075 Th, $^{13}\text{C}_1^{12}\text{C}_7\text{H}_{16}^{14}\text{N}_1^+$), TMT128L (128.12811564 Th, $^{13}\text{C}_1^{12}\text{C}_7\text{H}_{16}^{15}\text{N}_1^+$), TMT128H (128.13443559 Th, $^{13}\text{C}_2^{12}\text{C}_6\text{H}_{16}^{14}\text{N}_1^+$), TMT129L (129.13147048 Th, $^{13}\text{C}_2^{12}\text{C}_6\text{H}_{16}^{15}\text{N}_1^+$), TMT129H (129.13779043 Th, $^{13}\text{C}_3^{12}\text{C}_5\text{H}_{16}^{14}\text{N}_1^+$), TMT130L (130.13482532 Th, $^{13}\text{C}_3^{12}\text{C}_5\text{H}_{16}^{15}\text{N}_1^+$), TMT130H (130.14114527 Th, $^{13}\text{C}_4^{12}\text{C}_4\text{H}_{16}^{14}\text{N}_1^+$), and TMT131 (131.13818016 Th, $^{13}\text{C}_4^{12}\text{C}_4\text{H}_{16}^{15}\text{N}_1^+$). The labeled peptides were mixed in the 5|1|1|2|2|5|5|10|10|5 ratio following the mass tag order above. The sample was fractionated with reversed-phase chromatography at pH 12 (1 mm Xbridge column, Waters) into 25 fractions as previously described.³²

LC/MS/MS Analysis. Samples were dried *in vacuo* and resuspended in 0.05% TFA in water. 50% of the sample was injected into an Ultimate3000 nanoRLSC (Dionex) coupled to a Q Exactive (Thermo Scientific) or nanoAcquity (Waters) coupled to or LTQ-Orbitrap Elite (Thermo Scientific). Peptides were separated on custom 50 cm $\times$ 100 μ m (i.d.) reversed-phase columns (Reprosil) at 40 $^\circ\text{C}$. Gradient elution was performed from 2% acetonitrile to 40% acetonitrile in 0.1% formic acid over 2 h. The instrument was operated with a data dependent top 10 method. For the Q Exactive instruments MS spectra were acquired using 70,000 resolution and HCD scans at 35,000 or 70,000 resolution (at m/z 200). For the LTQ-Orbitrap Elite instrument MS spectra were acquired at 60,000 resolution and HCD scans at 30,000 and 60,000 resolution (at m/z 400). The MS ion targets were 3e6 for the Q Exactive and 1e6 for the LTQ-Orbitrap Elite. For MS2 spectra different ion target settings were tested (1e6, 2e5). The respective software versions were Tune 2.7 for the LTQ-Orbitrap Elite and Tune 2.2 for the Q Exactive and Xcalibur 2.7.

Peptide and Protein Identification. Mascot 2.2 (Matrix Science) was used for protein identification using a 10 ppm mass tolerance for peptide precursors and 20 mDa (HCD) mass tolerance for fragment ions. Carbamidomethylation of cysteine residues and TMT modification of lysine residues were set as fixed modifications and methionine oxidation; N-terminal acetylation of proteins and TMT modification of peptide N-termini were set as variable modifications. The search database consisted of a customized version of the IPI protein sequence database combined with a decoy version of this database created using a script supplied by Matrix Science. Unless stated otherwise, we accepted protein identifications as follows: (i) For single spectrum to sequence assignments, we required this assignment to be the best match *and* a minimum Mascot score of 31 *and* a 10 $\times$ difference of this assignment over the next best assignment. On the basis of these criteria, the decoy search results indicated <1% false discovery rate (FDR). (ii) For multiple spectrum to sequence assignments and with the same parameters, the decoy search results indicate <0.1% false

discovery rate. All identified proteins were quantified; FDR for quantified proteins was $\ll$ 0.1%.

Extraction of TMT Reporter Ion Signals. Reporter ion signals are extracted from the MS2 spectra using a modified version of the algorithm that is used to detect ions in the MS1 spectra. The modification is to reduce the minimum separation between ion signals. In the MS1 spectra we require a minimum intensity valley of 50%, the maximum intensity to separate ions. For the MS2 reporter ions the valley is 1%. This allows us to detect separate signals for ions that have higher levels of overlap (coalescence). These ions are analyzed by an algorithm, conceptually similar to the one described by Pachl et al.,³³ designed to find the most coherent set of reporter ions even in data where the calibration had shifted from the ideal. In these circumstances, because reporter ions are confined to a very limited m/z range (TMT range is 5.010 Th), they can be considered to have the same mass calibration offset, whereas their relative mass differences remain the same.

The algorithm works in several steps. The first collects ions in a wide tolerance (10 mDa) around the theoretical mass of each reporter ion. The mass differences (Seed Deltas) for each identified ion and the theoretical m/z of the reporter ions are stored.

Each Seed Delta is used to identify ions within a narrow tolerance (3.16 mDa) of this offset from the expected reporter ion m/z 's (modified m/z : m/z – Seed Delta). The tolerance of 3.16 mDa ensures that an ion cannot be simultaneously matched to two theoretical proximate reporter ions, and at the same time the tolerance is sufficiently wide for the correct matching still to occur in case a partial coalescence of two proximate reporter ions has taken place. From the detected ions within the tolerance range all possible reporter ion set combinations are created, and the best reporter ion cohort for a Seed Delta is selected using the cascade of criteria below, only passing to the next level if more than one entry exists in the current level: (1) the highest number of identified reporter ions, (2) the highest summed reporter ion area of all detected reporter ions (SRIA), (3) least spread from the Modified m/z 's.

Once reporter ion sets have been found for all Seed Deltas the maximum number of reporter ions found in a set is determined, and any set with more than two fewer ions than the maximum number is eliminated. The remaining sets are ranked by their SRIA, and the set with the highest SRIA is selected as the representative reporter ion set.

Peptide and Protein Quantification. Matching peptides were required to be unique for the identified protein. Reporter ion intensities were read from raw data using in-house software³⁴ and multiplied with ion accumulation times (the unit is milliseconds) to yield a measure proportional to the number of ions; this measure is referred to as ion area throughout the text. Peptide matched spectra were filtered according to the following criteria: mascot ion score >15, signal-to-background of the precursor ion >4, and signal-to-interference >0.5.^{34,35} Fold changes were corrected for isotope purity as described and adjusted for interference caused by coeluting nearly isobaric peaks as estimated by the signal-to-interference measure.³⁶ Protein quantification was derived from individual matching spectra using a sum-based bootstrap algorithm.³⁴ Relative protein abundances were generated on the basis of MS1 abundances.^{37,38} Briefly, XIC peaks were matched to identified peptides. The apex of the XIC peak was required to be within 30 s from the time of the MS2 event performed on the peptide precursor. The raw abundances of

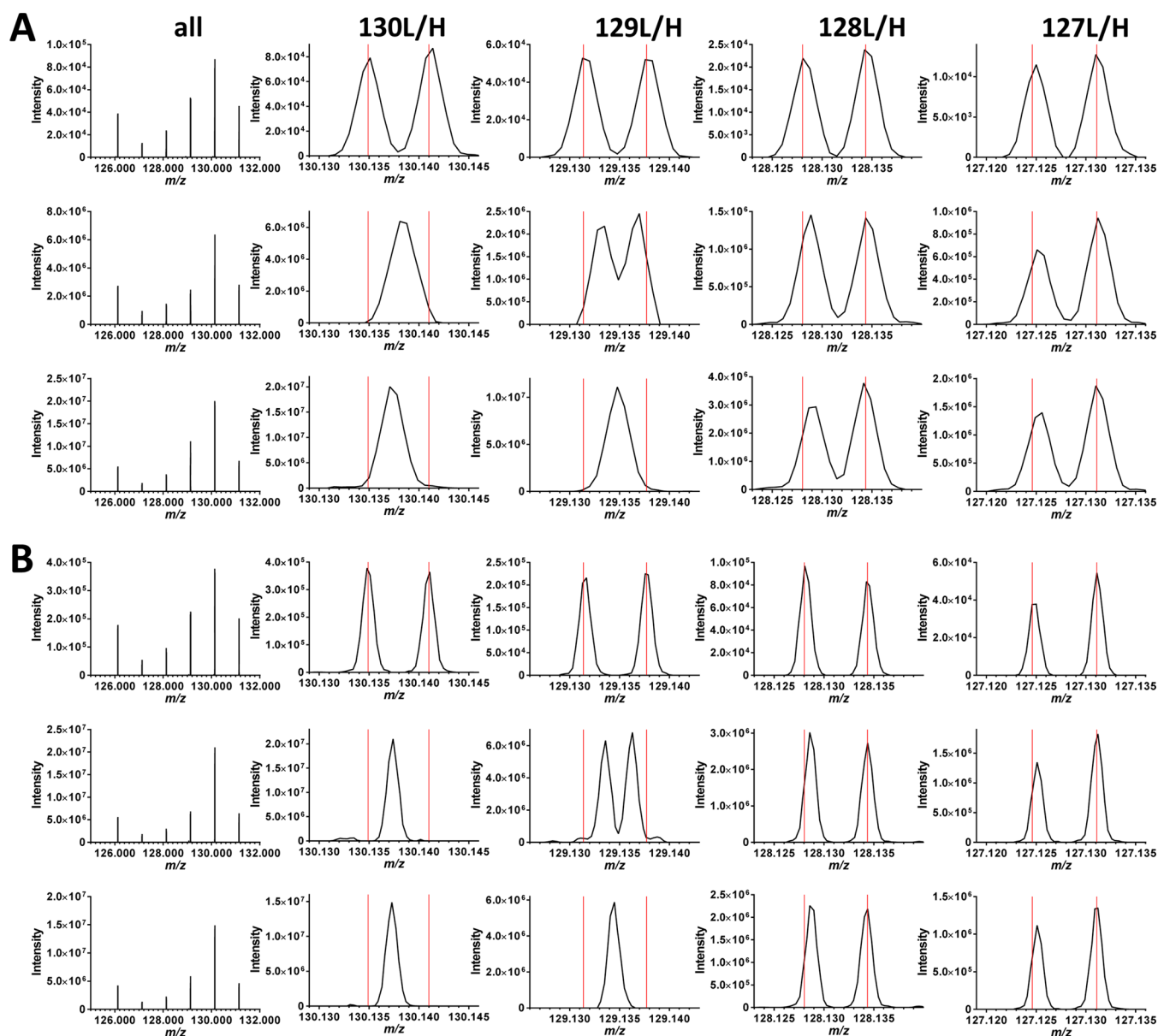


Figure 1. Examples of varying degrees of coalescence (A) for an experiment run at a resolution setting of 35,000 and an MSn ion target of $1e6$ on the Q Exactive. Upper panel: MS2 spectrum with no coalescence. Middle panel: MS2 spectrum with complete coalescence of TMT130L/H pair and partial coalescence of TMT129L/H pair. Lower panel: MS2 spectrum with complete coalescence of TMT130L/H and TMT129L/H pairs. (B) Identical examples of coalescence for an experiment run at a resolution setting of 70,000 and an MSn ion target of $1e6$ on the Q Exactive. Note that the reporter ion intensities are not corrected for ion accumulation times.

the XIC peaks of the peptides with identical sequences were summed (i.e., same sequence, but different charge states and/or different modifications), and the resulting single entity was referred to as a sequence. For each protein the 3 sequences with the highest raw XIC intensity from a given sample were selected and log 10 transformed. These values were then summed, and the mean was calculated. In cases where fewer than 3 sequences were associated, the mean was calculated using XIC intensities from either one or two sequences.

Statistical Analysis. The robustness of the TMT based protein quantification is directly related to how many spectra have been acquired for the protein; hence, the proteins were divided into bins according to how many spectra have been used for the quantification. The bins are constructed starting from proteins with the lowest number of spectra. Each bin consists of at least 300 proteins, and all proteins with the same

number of spectra are in the same bin. If there are less than 300 proteins with a certain number of spectra, the bin is filled up with proteins with a one plus higher number of spectra; this is repeated until the 300 value is reached. Additionally once each bin has been completed the remaining number of proteins (all with higher number of spectra) is counted; if this number is below 300, the remaining proteins are added to the last completed bin. The statistical significance of differences in protein fold changes was calculated by estimating the underlying normal distribution of the measurements and calculating the p -values for all measurement for a specific bin. Subsequently a correction for multiple hypothesis testing was performed on the full data set using Benjamini–Hochberg correction.

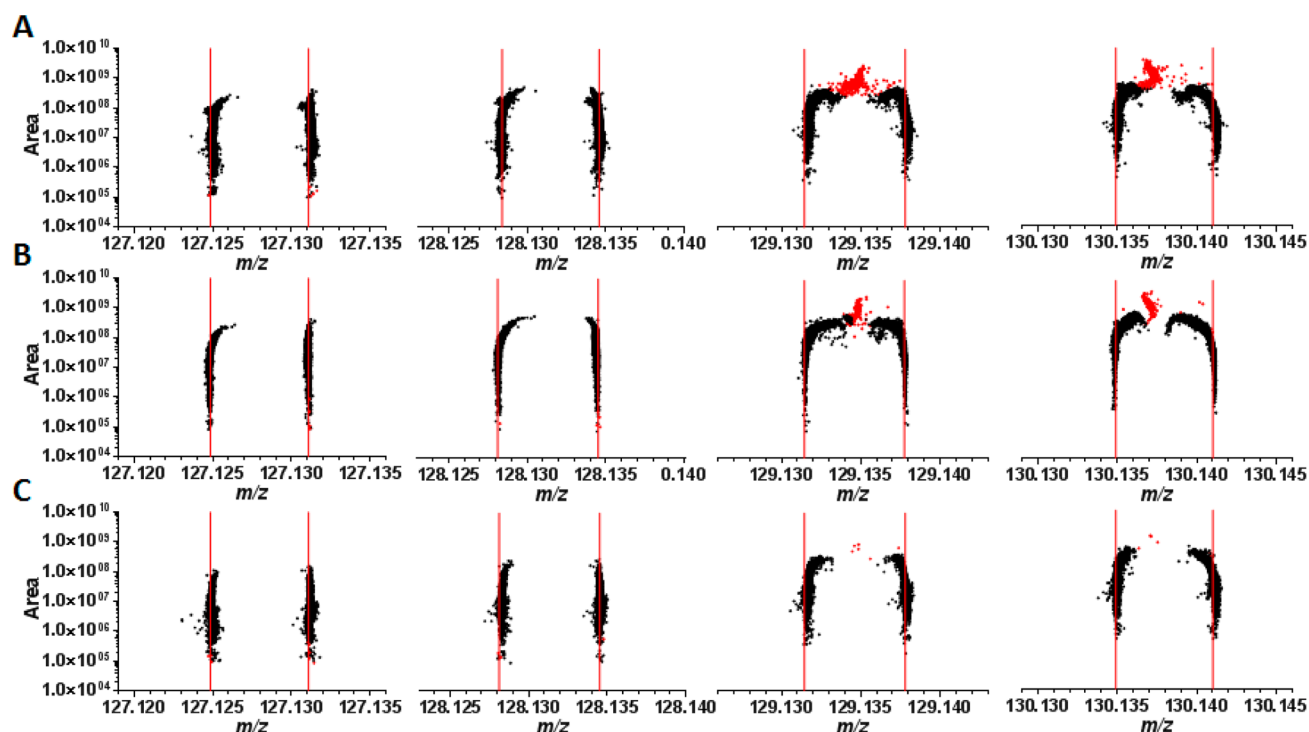


Figure 2. Scatter plots of reporter ion areas and mass deviations from theoretical values across all spectra from a given experiment. In cases where both reporter ion signals for a given 6 mDa separated pair were detected, both are marked as black dots. Cases where only one member of the pair was detected are marked as red dots. (A) For an experiment run at a resolution setting of 35,000 on the Q Exactive with an MSn ion target of $1e6$. (B) For an experiment run at a resolution setting of 70,000 on the Q Exactive with an MSn ion target of $1e6$. (C) For an experiment run at a resolution setting of 35,000 on the Q Exactive with an MSn ion target of $2e5$.

RESULTS AND DISCUSSION

In order to achieve the best analytical depth and spectra quality of complex proteomics samples, the MSn ion target of the Q Exactive instruments is typically set to $1e6$ ions, and the maximum ion injection time is set to be close to the transient time required for the desired resolution setting.^{39,40} This would correspond to using a maximum injection time of 60 and 120 ms when the resolution is set to 17,500 (transient time 64 ms) and 35,000 (transient time 128 ms), respectively.³⁹ These settings do not compromise the speed of the instrument due to the parallel filling and Orbitrap acquisition of the Q Exactive.³⁹ The net effect of the high MSn ion target will be that the maximum injection time is reached for almost all precursors selected for fragmentation without adversely affecting acquisition speed. We had previously shown that when working with samples labeled with neutron encoded TMT mass tags (such as TMT127L/H, TMT128L/H, TMT129L/H, and TMT130L/H, see Materials and Methods section) a resolution of 30,000 in MS2 at 400 m/z spectra is necessary to achieve baseline resolution of the proximate, 6 mDa spaced, reporter ion pairs on the Orbitrap Elite.²⁴ In order to evaluate the performance of the neutron encoded TMT labels on the Q Exactive instruments, we labeled 10 aliquots of K562 cell lysate with TMT10 and mixed at defined ratios (see Materials and Methods section) and fractionated the sample into 25 fractions, and individual fractions were analyzed on the Q Exactive. A resolution of 35,000 at 200 m/z (this corresponds to a resolution of ca. 50,000 at 130 m/z) was necessary in order to baseline resolve the four neutron encoded reporter ion pairs TMT127L/H, TMT128L/H, TMT129L/H, TMT130L/H, on the Q Exactive instrument (Figure 1A, upper panel). However, closer inspection revealed cases of partial, as well as full,

coalescence (Figure 1A, middle and lower panel) of the most abundant TMT130L/H and second to most abundant TMT129L/H reporter ion pairs, when using the standard MSn ion target setting for Q Exactive instruments of $1e6$ ions and a maximum acquisition time of 120 ms. When partial coalescence occurs a mass shift of the proximate reporter ions toward the center is observed; in the case of full coalescence the reporter ion doublets coalesce into a single peak. In order to fully characterize the extent of the coalescence when analyzing TMT10 labeled samples using different instrument settings as well as different instruments we performed experiments with different MS2 resolution settings (35,000 and 70,000 on Q Exactive instruments and 30,000 and 60,000 on the Orbitrap Elite instrument), with different MSn ion targets ($1e6$, $2e5$), and on three different instruments (two Q Exactives and an Orbitrap Elite), see Materials and Methods section.

The coalescence phenomenon has been extensively studied on FTICR instruments.^{29,31} A recent report has also demonstrated that it can occur when using Orbitrap instrumentation. Using two close in mass peptides Gorshkov et al. observed coalescence for the first time on an Orbitrap Elite instrument at an m/z value of 1060 Da, for two peaks spaced by 22 mDa and allowing fill times of up to 8 s in order to accumulate a sufficient number of ions.³⁰

The four neutron encoded pairs of TMT10 reporter ions (TMT127L/H, TMT128L/H, TMT129L/H, TMT130L/H) have a smaller spacing of 6 mDa, which should facilitate coalescence, (i.e., require less ions in the trap in order for coalescence to take place), but their m/z is much smaller (≈ 8 times) which according to theory should make it harder for them to coalesce (i.e., require more ions in the trap in order for coalescence to take place).²⁹ Additionally the relatively short

fixed injection time of 120 ms should make it harder to accumulate sufficient ion numbers for coalescence; hence, the appearance of coalescence was unexpected.

Fractions of the K562 sample were run using different ion target and resolution settings. According to FTMS theory,^{29–31} increasing the resolution just by increased transient time should not have had any mitigating effect on coalescence, but it should have made it more apparent whether the coalescence is complete or partial. Both experiments performed at 35.000 resolution and 70.000 resolution on the Q Exactive instrument, with a 1e6MSn ion target, had clear cases of full and partially coalesced proximate reporter ion peaks (Figure 1A,B). For the 35.000 resolution setting 485 out of 8798 TMT130L/H reporter ion pairs coalesced (i.e., only a single peak could be detected, see Materials and Methods section), and 554 out of 8798 TMT129L/H reporter ion pairs coalesced. For the 70.000 at 200 *m/z* resolution setting (this corresponds to a resolution of ca. 100.000 at 130 *m/z*), 230 out of 6372 TMT130L/H reporter ion pairs coalesced, and 253 out of 6372 TMT129L/H reporter ion pairs coalesced. As expected coalescence starts to occur for high abundance reporter ions peak pairs (Figure 2A,B). The strongest effect was observed for the TMT130L/H and TMT129L/H pairs, which were also the most and next to most abundant isobaric reporter ion pairs overall in the sample. The number of cases where only one peak could be detected instead of a pair was greater when a resolution of 35.000 was used which was most likely due to the fact that partially coalesced peaks can be better resolved using a higher resolution. The coalescence effect took place at a well-defined log 10 area value of the peaks, 8.8 for the experiment with the 35.000 resolution setting, meaning that there were only fully coalesced peaks with a log 10 area value of greater than 8.8 (Figure S-1A,B in the Supporting Information). The corresponding value for the experiment with the 70.000 resolution setting was 8.9 (Figure S-1C,D in the Supporting Information); the small difference of 0.1 is most likely due to the enhanced sharpness of the peaks at higher resolution, and consequently higher area of the peaks for the same number of ions. In the experiment using an MSn ion target of 2e5 almost no coalescence was observed; only 8 TMT130L/H and 7 TMT129L/H reporter ion pairs were found to be fully coalesced (Figure 2C). The number of proteins identified remained virtually the same when reducing the MSn ion target setting from 1e6 (2753 proteins) to 2e5 (2737 proteins) while retaining the same resolution of 35.000 (Table 1). Increasing the resolution to 70.000 had an adverse effect on protein identification (2368 proteins, Table 1).

Table 1. Numbers of Proteins Identified and Quantified on Different Instruments with Different Resolution and MSn Ion Target Settings

instrument	MSn ion target	resolution	proteins identified
Q Exactive (1)	1.00×10^6	35 000	2753
Q Exactive (1)	1.00×10^6	70 000	2368
Q Exactive (1)	2.00×10^5	35 000	2737
Q Exactive (2)	1.00×10^6	35 000	2731
Q Exactive (2)	1.00×10^6	70 000	2456
Q Exactive (2)	2.00×10^5	35 000	2657
Orbitrap Elite	1.00×10^6	30 000	2166
Orbitrap Elite	1.00×10^6	60 000	1756
Orbitrap Elite	2.00×10^5	30 000	2252

Coalescence of reporter ion pairs will lead to improper quantification of respective peptide spectra, but ultimately also the quantification of the corresponding protein. The area of the coalesced peak will be interpreted as only one of the two possible TMT signal values, depending on its mass shift (see Materials and Methods), and the other signal will be counted as having an area of zero. Since coalescence will affect the spectra with the strongest TMT signals, it will have a strong effect on protein quantification when using common approaches such as sum of intensities⁴¹ or in our case sum of areas, where most weight is automatically given to the strongest TMT signals. When calculating the TMT130L to TMT131 log 2 ratio on the protein level in the experiment performed at 35.000 resolution with MSn ion target set to 1e6, it becomes apparent that there is a systematic bias for high abundant proteins (Figure 3A). In case of coalescence the coalesced peak is in most cases interpreted as a TMT130L signal, and hence the overall TMT130L area of the protein becomes inflated which results in fold changes of greater than the expected 1 on the log 2 scale. Conversely, for the TMT130H to TMT131 ratio the fold change is often significantly below the log 2 value of 1, because in case of coalescence the TMT130H gets assigned a value of zero which leads to an underestimation of the total TMT130H area of the protein compared to TMT131 and skews the quantification (Figure 3B). The fold change bias of the abundant proteins affected by coalescence makes them appear as significantly regulated within our statistical framework (see Materials and Methods section) which is what led us to discover the coalescence phenomenon when analyzing relevant biological experiments. In the experiment where the 2e5MSn ion target was used the quantification bias is absent (Figure 3C,D), since virtually no coalescence occurred in this experiment.

The coalescence effect was less prominent on a second Q Exactive instrument when using the 1e6MSn ion target (Figure S-2 in the Supporting Information), 48 cases of full coalescence out of 11820 TMT130L/H pairs, for the 35.000 resolution experiment and 45 out of 5785 TMT130L/H pairs, for the 70.000 resolution experiment. In both experiments the threshold for complete coalescence was a log 10 area value of 9, marginally higher than on the first Q Exactive instrument. The bias in protein quantification was consequently also present, but to a lesser extent (Figure S-3 in the Supporting Information). Only one case of full coalescence occurred when using the 2e5MSn ion target, (Figure S-2 in the Supporting Information). The protein identification trend was similar to the trend on the first Q Exactive with near identical numbers identified at 35.000 resolution with MSn ion target settings of 1e6 or 2e5 and a reduced number of proteins identified when using the 70.000 resolution and 1e6MSn ion target setting (Table 1).

The Orbitrap Elite had the least coalescence, with only 14 cases of full coalescence out of 8660 TMT130L/H pairs, for the 30.000 resolution experiment and zero out of 8216 TMT130L/H pairs, for the 60.000 resolution experiment; partial coalescence was present with both settings (Figure S-4 in the Supporting Information). The coalescence effects were completely absent when using the 2e5MSn ion target setting (Figure S-4 in the Supporting Information). Consequently, systematic effects on protein quantification were not observed either with the high or the low MSn ion target setting (Figure S-5 in the Supporting Information). Changes in ion target and resolution settings led to similar effects on protein identification

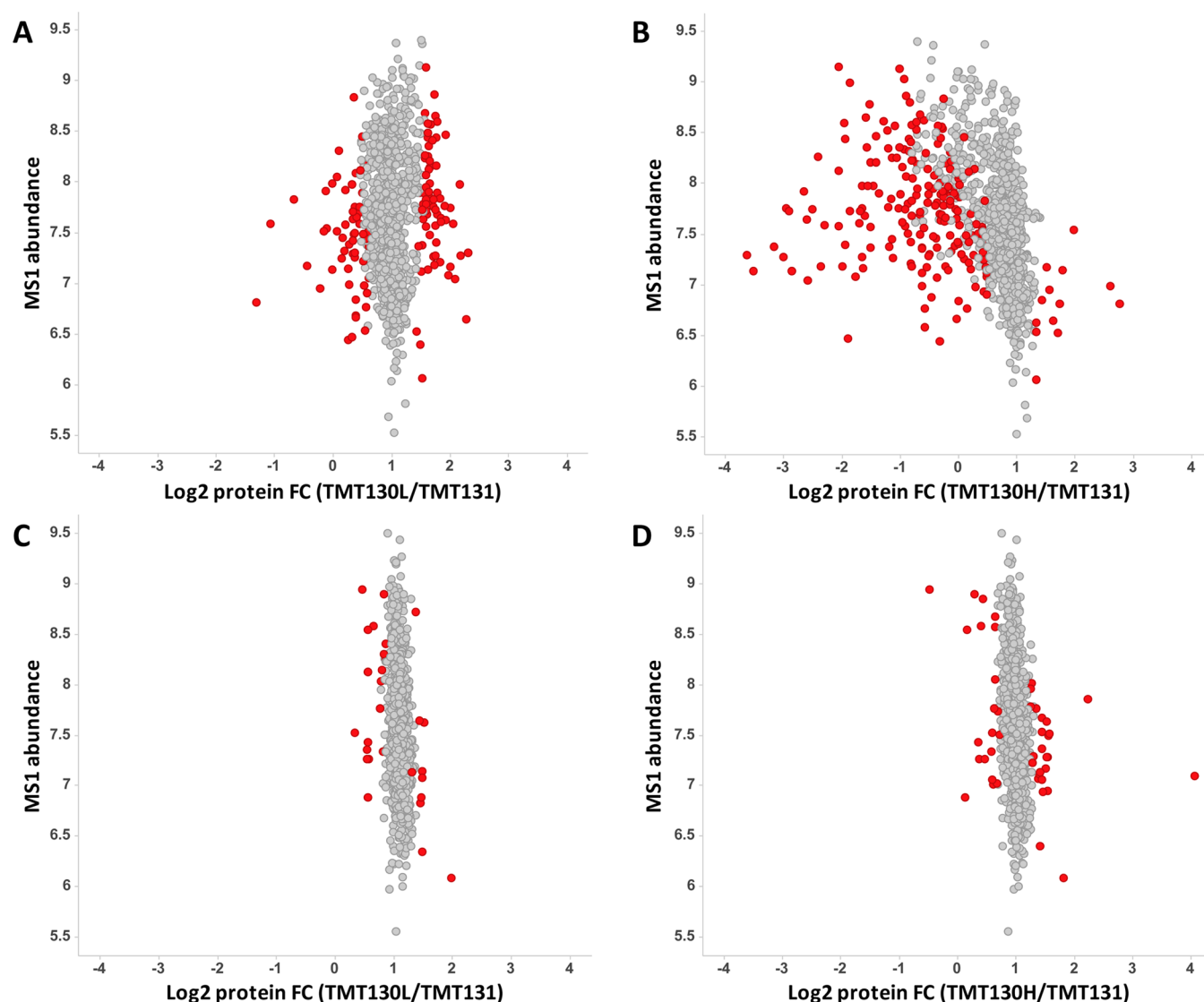


Figure 3. Measured log 2 protein fold changes, red color signifies fold changes that have a p -value of below 0.05. (A) TMT130L versus TMT131 (1 Thomson spaced reporter ions) and (B) TMT130H versus TMT131 (1 Thomson spaced reporter ions) for an experiment run at a resolution setting of 35,000 on the Q Exactive with an MSn ion target of 1e6; (C) TMT130L versus TMT131 and (D) TMT130H versus TMT131 for an experiment run at a resolution setting of 35,000 on the Q Exactive with an MSn ion target of 2e5.

rates as with the Q Exactive experiments. Similar numbers of proteins were identified at 30,000 resolution with MSn ion target settings of 1e6 or 2e5, respectively, and a smaller number of proteins identified when using the 60,000 resolution and 1e6MSn ion target setting (Table 1). The negligible occurrence of coalescence on the Orbitrap Elite is most likely due to the compact, high-field Orbitrap analyzer employed in this instrument. It is well-known that on FTICR instruments the tendency of two ion packets to coalesce drops with increasing magnetic field strength.³¹ Consequently, on the Orbitrap mass spectrometers the propensity for coalescence should drop with increasing electric field strength which is achieved in the Orbitrap Elite analyzer. Additionally, it should be noted that the generally used MSn ion target setting for Orbitrap Elite is well below 1e6;^{28,42} thus, coalescence should in principle never be a problem on these instruments. The above illustrates that using the Q Exactive instruments for commonly accepted 1e6MSn ion target setting in combination with a maximum allowed injection time of 120 ms can lead to ion coalescence of

proximate TMT10 reporter ion pairs (up to 6% of TMT130L/H pairs in our sample depending on the instrument used) when analyzing complex proteomics samples. This in turn has an adverse effect on protein quantification. Operating the instruments with a 2e5MSn ion target setting completely remedies the problem without any negative effect on protein identification or the precision of quantification accuracy. An alternative approach not pursued in this study would be to continue using the high MSn ion target of 1e6 and subsequently filter away all MS2 spectra where coalescence occurred. However, since the very simple solution of lowering the MSn ion target had no adverse effects and solved the problem we chose not to consider the filtering strategy. The most fail-safe strategy for dealing with coalescence of the neutron encoded TMT10 reporter ions would be a method for automatic on-the-fly correction of the MSn ion target value if any of the neutron encoded reporter ion areas exceed the instrument specific coalescence threshold, and reacquisition of the MS2 spectrum.

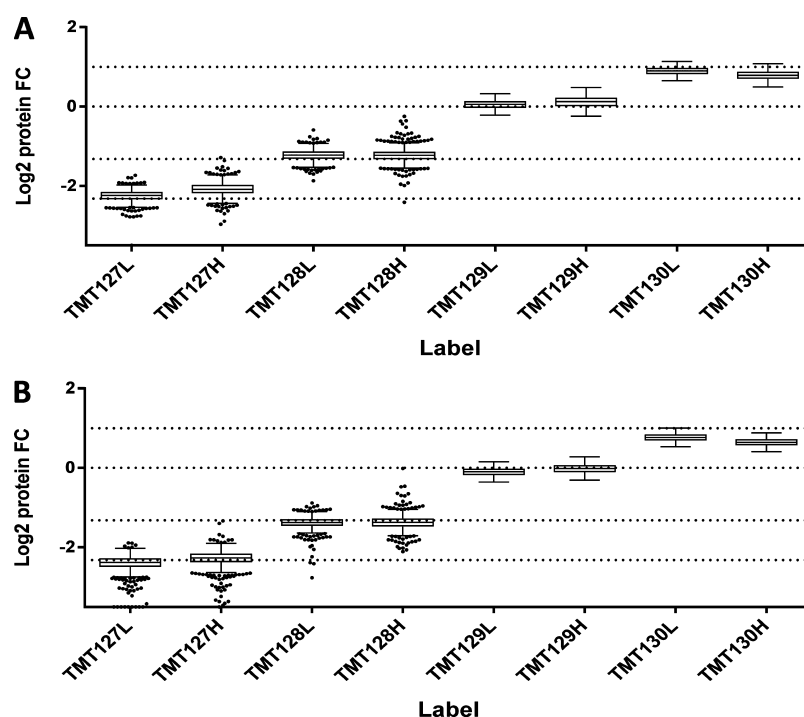


Figure 4. Box plot representation of the measured log₂ protein fold changes of the different TMT labels on the *x*-axis versus the TMT131 label. Dotted lines represent the expected log₂ fold change values. (A) For an experiment run at a resolution setting of 35,000 on the Q Exactive with an MSn ion target of 2e5. (B) For an experiment run at a resolution setting of 30,000 on the Orbitrap Elite with an MSn ion target of 2e5.

Finally we compared the quantification quality when using TMT10 on the first Q Exactive for the experiment run with a resolution setting of 35,000 and MSn ion target of 2e5 (Figure 4A), with the experiment run with a resolution setting of 30,000 and MSn ion target of 2e5 on the Orbitrap Elite (Figure 4B). Both the means and standard deviations of the quantified proteins differ very little between the two instruments (Table S-1 in the Supporting Information) and agree well with the mixed ratios. The average deviation for the different conditions compared on the Q Exactive is 0.13 and 0.11 on the Orbitrap Elite. This demonstrates that quantification with TMT10 isobaric mass tags can be achieved on the Q Exactive instruments with comparable precision and accuracy to the Orbitrap Elite.

CONCLUSIONS

We have observed a significant occurrence of coalescence of proximate (6 mDa spaced) TMT10 reporter ions in complex samples when analyzed on Q Exactive instruments using commonly accepted instrument settings. Up to 6% depending on the instrument of the most abundant proximate reporter ion pair fully coalesced into a single entity in MS₂ spectra. This had an adverse effect on peptide as well as protein quantification. Reduction of the MSn ion target setting from 1e6 to 2e5 prevents coalescence from occurring without negative effect on protein identification or quantification. Our data suggest that ion coalescence can occur in real life proteomic samples and might also impair the mass accuracy and precision of quantification of intact peptide ions, in particular for neutron-encoded MS1 quantification strategies.⁴³ Under conditions where coalescence is avoided, we demonstrated good accuracy and precision of quantification with 10-plex TMT tags on Q Exactive instruments.

ASSOCIATED CONTENT

Supporting Information

Additional information as noted in text. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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REFERENCES

- (1) Bantscheff, M.; Lemeer, S.; Savitski, M. M.; Kuster, B. *Anal. Bioanal. Chem.* **2012**, 404, 939–965.
- (2) Bantscheff, M.; Schirle, M.; Sweetman, G.; Rick, J.; Kuster, B. *Anal. Bioanal. Chem.* **2007**, 389, 1017–1031.
- (3) Beck, M.; Schmidt, A.; Malmstroem, J.; Claassen, M.; Ori, A.; Szymborska, A.; Herzog, F.; Rinner, O.; Ellenberg, J.; Aebersold, R. *Mol. Syst. Biol.* **2011**, 7, 549.
- (4) Nagaraj, N.; Wisniewski, J. R.; Geiger, T.; Cox, J.; Kircher, M.; Kelso, J.; Paabo, S.; Mann, M. *Mol. Syst. Biol.* **2011**, 7, 548.
- (5) Schwanhauser, B.; Busse, D.; Li, N.; Dittmar, G.; Schuchhardt, J.; Wolf, J.; Chen, W.; Selbach, M. *Nature* **2011**, 473, 337–342.
- (6) Pflieger, D.; Junger, M. A.; Muller, M.; Rinner, O.; Lee, H.; Gehrig, P. M.; Gstaiger, M.; Aebersold, R. *Mol. Cell. Proteomics* **2008**, 7, 326–346.

- (7) Przybylski, C.; Junger, M. A.; Aubertin, J.; Radvanyi, F.; Aebersold, R.; Pflieger, D. *J. Proteome Res.* **2010**, *9*, 5118–5132.
- (8) Iwai, L. K.; Benoist, C.; Mathis, D.; White, F. M. *J. Proteome Res.* **2010**, *9*, 3135–3145.
- (9) Olsen, J. V.; Blagoev, B.; Gnäd, F.; Macek, B.; Kumar, C.; Mortensen, P.; Mann, M. *Cell* **2006**, *127*, 635–648.
- (10) Rubbi, L.; Titz, B.; Brown, L.; Galvan, E.; Komisopoulou, E.; Chen, S. S.; Low, T.; Tahmasian, M.; Skaggs, B.; Muschen, M.; Pellegrini, M.; Graeber, T. G. *Sci. Signaling* **2011**, *4*, ra18.
- (11) Bantscheff, M.; Eberhard, D.; Abraham, Y.; Bastuck, S.; Boesche, M.; Hobson, S.; Mathieson, T.; Perrin, J.; Raida, M.; Rau, C.; Reader, V.; Sweetman, G.; Bauer, A.; Bouwmeester, T.; Hopf, C.; Kruse, U.; Neubauer, G.; Ramsden, N.; Rick, J.; Kuster, B.; Drewes, G. *Nat. Biotechnol.* **2007**, *25*, 1035–1044.
- (12) Bantscheff, M.; Hopf, C.; Savitski, M. M.; Dittmann, A.; Grandi, P.; Michon, A. M.; Schlegl, J.; Abraham, Y.; Becher, I.; Bergamini, G.; Boesche, M.; Delling, M.; Dumpelfeld, B.; Eberhard, D.; Huthmacher, C.; Mathieson, T.; PoECKel, D.; Reader, V.; Strunk, K.; Sweetman, G.; Kruse, U.; Neubauer, G.; Ramsden, N. G.; Drewes, G. *Nat. Biotechnol.* **2011**, *29*, 255–265.
- (13) Bantscheff, M.; Scholten, A.; Heck, A. J. *Drug Discov. Today* **2009**, *14*, 1021–1029.
- (14) Patricelli, M. P.; Nomanbhoy, T. K.; Wu, J.; Brown, H.; Zhou, D.; Zhang, J.; Jagannathan, S.; Aban, A.; Okerberg, E.; Herring, C.; Nordin, B.; Weissig, H.; Yang, Q.; Lee, J. D.; Gray, N. S.; Kozarich, J. W. *Chem. Biol.* **2011**, *18*, 699–710.
- (15) Remsing Rix, L. L.; Rix, U.; Colinge, J.; Hantschel, O.; Bennett, K. L.; Stranzl, T.; Muller, A.; Baumgartner, C.; Valent, P.; Augustin, M.; Till, J. H.; Superti-Furga, G. *Leukemia* **2009**, *23*, 477–485.
- (16) Afkarian, M.; Bhasin, M.; Dillon, S. T.; Guerrero, M. C.; Nelson, R. G.; Knowler, W. C.; Thadhani, R.; Libermann, T. A. *Mol. Cell. Proteomics* **2010**, *9*, 2195–2204.
- (17) Chao, T. C.; Hansmeier, N.; Halden, R. U. *J. Proteomics* **2010**, *73*, 1641–1646.
- (18) Schiess, R.; Wollscheid, B.; Aebersold, R. *Mol. Oncol.* **2009**, *3*, 33–44.
- (19) Turtoi, A.; Dumont, B.; Greffe, Y.; Blomme, A.; Mazzucchelli, G.; Delvenne, P.; Mutijima, E. N.; Lifrange, E.; De Pauw, E.; Castronovo, V. *J. Proteome Res.* **2011**, *10*, 3160–3182.
- (20) Ye, H.; Sun, L.; Huang, X.; Zhang, P.; Zhao, X. *Mol. Cell. Biochem.* **2010**, *343*, 91–99.
- (21) Thompson, A.; Schafer, J.; Kuhn, K.; Kienle, S.; Schwarz, J.; Schmidt, G.; Neumann, T.; Johnstone, R.; Mohammed, A. K.; Hamon, C. *Anal. Chem.* **2003**, *75*, 1895–1904.
- (22) Ross, P. L.; Huang, Y. N.; Marchese, J. N.; Williamson, B.; Parker, K.; Hattan, S.; Khainovski, N.; Pillai, S.; Dey, S.; Daniels, S.; Purkayastha, S.; Juhasz, P.; Martin, S.; Bartlett-Jones, M.; He, F.; Jacobson, A.; Pappin, D. J. *Mol. Cell. Proteomics* **2004**, *3*, 1154–1169.
- (23) Dayon, L.; Hainard, A.; Licker, V.; Turck, N.; Kuhn, K.; Hochstrasser, D. F.; Burkhard, P. R.; Sanchez, J. C. *Anal. Chem.* **2008**, *80*, 2921–2931.
- (24) Werner, T.; Becher, I.; Sweetman, G.; Doce, C.; Savitski, M. M.; Bantscheff, M. *Anal. Chem.* **2012**, *84*, 7188–7194.
- (25) McAlister, G. C.; Huttlin, E. L.; Haas, W.; Ting, L.; Jedrychowski, M. P.; Rogers, J. C.; Kuhn, K.; Pike, I.; Grothe, R. A.; Blethrow, J. D.; Gygi, S. P. *Anal. Chem.* **2012**, *84*, 7469–7478.
- (26) Hebert, A. S.; Merrill, A. E.; Bailey, D. J.; Still, A. J.; Westphall, M. S.; Strieter, E. R.; Pagliarini, D. J.; Coon, J. J. *Nat. Methods* **2013**, *10*, 332–334.
- (27) Hebert, A. S.; Merrill, A. E.; Stefely, J. A.; Bailey, D. J.; Wenger, C. D.; Westphall, M. S.; Pagliarini, D. J.; Coon, J. J. *Mol. Cell. Proteomics* **2013**, *12*, 3360–3369.
- (28) Michalski, A.; Damoc, E.; Lange, O.; Denisov, E.; Nolting, D.; Müller, M.; Viner, R.; Schwartz, J.; Remes, P.; Belford, M.; Dunyach, J.; Cox, J.; Horning, S.; Mann, M.; Makarov, A. *Mol. Cell. Proteomics* **2012**, *11*, DOI: 10.1074/mcp.O1111.013698.
- (29) Boldin, I. A.; Nikolaev, E. N. *Rapid Commun. Mass Spectrom.* **2009**, *23*, 3213–3219.
- (30) Gorshkov, M. V.; Fornelli, L.; Tsybin, Y. O. *Rapid Commun. Mass Spectrom.* **2012**, *26*, 1711–1717.
- (31) Scigelova, M.; Hornshaw, M.; Giannakopoulos, A.; Makarov, A. *Mol. Cell. Proteomics* **2011**, *10*, M111.009431.
- (32) Kruse, U.; Pallasch, C. P.; Bantscheff, M.; Eberhard, D.; Frenzel, L.; Ghidelli, S.; Maier, S. K.; Werner, T.; Wendtner, C. M.; Drewes, G. *Leukemia* **2011**, *25*, 89–100.
- (33) Pahl, F.; Fellenberg, K.; Wagner, C.; Kuster, B. *Proteomics* **2012**, *12*, 1328–1332.
- (34) Savitski, M. M.; Sweetman, G.; Askenazi, M.; Marto, J. A.; Lang, M.; Zinn, N.; Bantscheff, M. *Anal. Chem.* **2011**, *83*, 8959–8967.
- (35) Savitski, M. M.; Fischer, F.; Mathieson, T.; Sweetman, G.; Lang, M.; Bantscheff, M. *J. Am. Soc. Mass Spectrom.* **2010**, *21*, 1668–1679.
- (36) Savitski, M. M.; Mathieson, T.; Zinn, N.; Sweetman, G.; Doce, C.; Becher, I.; Pahl, F.; Kuster, B.; Bantscheff, M. *J. Proteome Res.* **2013**, *12*, 3586–3598.
- (37) Silva, J. C.; Gorenstein, M. V.; Li, G. Z.; Vissers, J. P.; Geromanos, S. J. *Mol. Cell. Proteomics* **2006**, *5*, 144–156.
- (38) Becher, I.; Savitski, M. M.; Savitski, M. F.; Hopf, C.; Bantscheff, M.; Drewes, G. *ACS Chem. Biol.* **2013**, *8*, 599–607.
- (39) Kelstrup, C. D.; Young, C.; Lavalley, R.; Nielsen, M. L.; Olsen, J. V. *J. Proteome Res.* **2012**, *11*, 3487–3497.
- (40) Wiśniewski, J. R.; Duś, K.; Mann, M. *Proteomics: Clin. Appl.* **2013**, *7*, 225–233.
- (41) Boehm, A. M.; Altmann, D.; Sickmann, A.; Falk, M. *BMC Bioinf.* **2007**, *8*, 214.
- (42) Pahl, F.; Ruprecht, B.; Lemeer, S.; Kuster, B. *Proteomics* **2013**, *13*, 2552–2562.
- (43) Ulbrich, A.; Merrill, A. E.; Hebert, A. S.; Westphall, M. S.; Keller, M. P.; Attie, A. D.; Coon, J. J. *J. Am. Soc. Mass Spectrom.* **2014**, *25*, 6–9.